

PII Detection and Masking

Internship Coding Assessment Report

Repository: <https://github.com/AbdullahShheik/PII-masking>

All training scripts, preprocessing code, and datasets used in this project are available in the repository.

1 Task Overview

The objective of this assessment was to build a system that detects and masks Personally Identifiable Information (PII) in text. Two entity types were considered:

- Person names (PER)
- Email addresses (EMAIL)

Two approaches were evaluated:

- Fine-tuned transformer-based NER model
- Zero-shot LLM-based PII masking

2 Dataset and Preprocessing

The base dataset used was the WikiNeural dataset containing:

- 28,516 training samples
- 3,650 test samples

Since the dataset only contains person entities, synthetic emails were generated by combining extracted names with common domains such as `gmail.com`, `yahoo.com`, `outlook.com`, and `hotmail.com`. Approximately **40% of training samples** were augmented with emails.

Two evaluation setups were created:

- **Scenario 1:** Synthetic emails added to the test data.
- **Scenario 2:** Independent test set built from external Kaggle email datasets.

Scenario 2 was used for final evaluation to better measure generalization.

3 Model Training

A **DeBERTa-v3-small** transformer was fine-tuned for token classification using BIO tagging.

The dataset is highly imbalanced since most tokens belong to the 0 label. The following strategies were explored:

- Baseline fine-tuning
- Early stopping
- Weighted loss for entity tokens
- Freezing lower encoder layers

Weighted loss was used to increase the importance of PER and EMAIL tokens.

4 Model Evaluation

4.1 Scenario 1: Augmented Test Data

When evaluated on synthetic test data generated using the same patterns as training, the model achieved near-perfect performance.

These results are expected since the training and test distributions are similar.

Model	PER F1	EMAIL F1	Overall F1
Baseline	0.989	1.000	0.985
Early Stop + Weighted	0.990	1.000	0.986
Frozen Layers	0.986	1.000	0.979

4.2 Scenario 2: Independent Test Dataset

A more realistic evaluation was performed using an independent dataset with diverse real-world email formats. The final selected model was the **Weighted Loss + Early Stopping model**.

Entity	Accuracy	Precision	Recall	F1	FPR	FNR
PER	0.9979	0.9911	1.0000	0.9955	0.0028	0.0000
EMAIL	0.9503	0.9975	0.5785	0.7323	0.0002	0.4215

Person detection achieves near-perfect performance. Email detection shows high precision but lower recall due to the wide variety of real-world email formats.

5 LLM Baseline

A zero-shot LLM approach was also evaluated.

5.1 Pipeline 1: Entity Extraction + Programmatic Masking

The LLM extracts entities in JSON format and masking is applied programmatically.

Metric	PER	EMAIL
Accuracy	0.884	0.898
Precision	0.446	0.236
Recall	0.618	0.610
F1	0.518	0.341

5.2 Pipeline 2: Direct Masking

The LLM directly outputs masked text without explicit entity extraction. This approach performed worse because detection and masking occur simultaneously, increasing errors.

6 Error Analysis and Improvement Suggestions

6.1 Error Analysis

The transformer model performs extremely well for person names but has lower recall for emails. This is mainly due to limited synthetic email patterns during training compared to the diversity of real-world formats.

The LLM pipelines show different issues. The extraction pipeline sometimes produces incorrect spans, while the direct masking approach occasionally masks incorrect tokens or misses entities.

6.2 Potential Improvements

Performance could be improved by adding more diverse email formats to the training data. Another potential improvement is adversarial training, where small perturbations are added to input embeddings during training to improve robustness. This was not explored due to time and computational constraints.

6.3 Strengths and Weaknesses

The transformer model provides high precision and reliable detection when sufficient labeled data is available. However, its performance depends on training data diversity.

The LLM approach requires no training and is flexible across tasks, but produced lower precision and less consistent masking in this evaluation.